

Kevin Chhim

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EDUCATION

University of California, Irvine

B.S in Computer Engineering, Concentration in Embedded Systems

Irvine, CA

Expected: June 2027

- Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Computer Architecture, Operating Systems, Computer Networks, Discrete-Time Signals & Systems, Databases, VLSI, AI/ML

PROJECTS

UnderWatch – Local AI Surveillance System

Mar. 2026 – Present

Skills: Python, C++, Edge Impulse, Arduino, Computer Vision, ML

Irvine, CA

- Built a local AI surveillance system on a **dual-processor architecture**, coordinating **ML inference** on the **Qualcomm QRB2210** MPU with real-time servo and buzzer control on the **STM32 MCU** running **Zephyr RTOS**
- Trained and deployed a fine-tuned **Edge Impulse** image classification model achieving **~93% accuracy** with confidence scoring for **fall detection**
- Leveraged **Qualcomm** silicon's edge AI capabilities to run inference entirely **on-device**, eliminating cloud dependency and achieving **real-time detection latency under 200ms**

FPV Drone – Research & Design

Sep. 2025 – Present

Skills: ArduPilot, Mission Planner, STM32, Raspberry Pi, FPV Systems

Irvine, CA

- Built a **3.5–4.5” modular FPV quadcopter** from a bare frame, integrating motors, ESCs, **STM32-based flight controller**, GPS, radio receiver, and FPV camera system
- Performed full **electrical assembly** including **power distribution** (XT30/XT60), soldering signal lines, and managing **EMI** between video and radio subsystems
- Troubleshoot **firmware** and connectivity issues (**ExpressLRS** linking, motor desync) and validated flight using RadioMaster controller and FPV monitor

SmartTart – IoT-Based Smart Toaster

Apr. 2025 – June 2025

Skills: C++, Arduino, ESP32, IoT, Google Gemini API

Irvine, CA

- Programmed **ESP32** microcontroller to control toasting cycles using **sensor input** and **API integration**
- Integrated an **infrared temperature sensor** and **accelerometer** for real-time heat and motion monitoring, with **relay-based heater control** for full device interactivity
- Built a **local web server** to receive **API POST requests** and set user profiles via **natural language input**
- Collaborated in a multidisciplinary team to build **full-stack** functionality within a **36-hour hardware hackathon**

BCN3D Moveo – Open Source Robotic Arm Design

Sep. 2024 – June 2025

Skills: SolidWorks, Arduino, C++

Irvine, CA

- Developed **firmware** for the robotic arm using the **Arduino** platform, enabling **precise control** of motors and sensors
- Designed a **power management system** ensuring stable voltage and current supply for reliable performance
- Collaborated on the **mechanical design and assembly** using **SolidWorks**, optimizing the arm's structural integrity and movement

EXPERIENCE

Tech Director

Jun. 2025 – Present

Engineering Student Council (ESC)

Irvine, CA

- Built a **room booking platform** from scratch in **Next.js/TypeScript/Tailwind** with a **custom calendar** (no libraries), replacing scattered manual sign-ups
- Added **Firestore OAuth** so only engineering students could book; **backend logic** prevented **double-bookings**

Embedded Systems Engineer

Sep. 2025 – Present

Micromouse (MM)

Irvine, CA

- Developing **flood-fill maze-solving algorithms** in **C/C++** on an **STM32 microcontroller**, enabling **real-time autonomous navigation** and dynamic path optimization
- Applied **circuit design**, **PCB design**, soldering, and **hardware debugging** to optimize robotic subsystems in a collaborative engineering environment

SKILLS

Programming: C, C++, Python, Java, JavaScript, Rust, Assembly (x86-64, MIPS), Verilog, SystemVerilog, VHDL, MATLAB
Hardware: Altium, KiCad, STM32, ESP32, ARM, DSP, Raspberry Pi, Arduino, FPGA, Soldering, Logic Analyzer, SolidWorks
Protocols & Interfaces: UART, SPI, I2C, PWM, GPIO, JTAG, ExpressLRS
Development Tools: Git, VS Code, STM32CubeIDE, Vivado, ModelSim, GDB, OpenOCD, Linux, RTOS